1.06 Quadratics: Introduction

Worked Solutions

1 a)

$x^2 + 14x + 45 = x^2 + 9x + 5x + 45$

$x(x + 9) + 5(x + 9) = (x + 9)(x + 5)$

1 b)

$x^2 + 3x - 18 = x^2 - 3x + 6x - 18$

$= x(x - 3) + 6(x - 3) = (x - 3)(x + 6)$

1 c)

$x^2 - 7x = x(x - 7)$

1 d)

$2x^2 + 11x - 6 = 2x^2 + 12x - x - 6$

$= 2x(x + 6) - 1(x + 6) = (x + 6)(2x - 1)$

1 e)

$2x^2 - 17x + 35 = 2x^2 - 10x - 7x + 35$

$= 2x(x - 5) - 7(x - 5) = (x - 5)(2x - 7)$

1 f)

$8x^2 - 14x + 3 = 8x^2 - 12x - 2x + 3$

$= 4x(2x - 3) - 1(2x - 3) = (2x - 3)(4x - 1)$

1 g)

$6x^2 + 17x + 5 = 6x^2 + 2x + 15x + 5$

$= 2x(3x + 1) + 5(3x + 1) = (3x + 1)(2x + 5)$

1 h)

$-2x^2 - x + 3 = -2x^2 - 3x + 2x + 3$

$-x(2x + 3) + 1(2x + 3) = (2x + 3)(1 - x)$

1 i)

$x^2 - 9 = x^2 + 3x - 3x - 9$

$= x(x + 3) - 3(x + 3) = (x - 3)(x + 3)$

1 j)

$4x^2 - 5 = 4x^2 + 2\sqrt{5}x - 2\sqrt{5}x - 5$

$= 2x(2x + \sqrt{5}) - \sqrt{5}(2x + \sqrt{5})$

$= (2x - \sqrt{5})(2x + \sqrt{5})$

2 a)

$x^2 - a^2 = (x - a)(x + a)$

2 b)

$(ax)^2 - b^2 = (ax - b)(ax + b)$

2 c)

$x^2 - a = (x - \sqrt{a})(x + \sqrt{a})$

3 a)

$8(x + 2) - x(x + 2) = (x + 2)(8 - x)$

3 b)

$-(x - 2) + 2x(x - 2) = (x - 2)(2x - 1)$

3 c)

$12(x - 5)^2 + (x - 5) = (x - 5)[12(x - 5) + 1]$

$= (x - 5)[12x - 60 + 1] = (x - 5)(12x - 59)$

3 d)

$(x - 7)^2 - 4 = x^2 - 14x + 49 - 4$

$x^2 - 14x + 45 = x^2 - 9x - 5x + 45$

$= x(x - 9) - 5(x - 9) = (x - 5)(x - 9)$

3 e)

$\frac{2}{3}(3x - 10)^2 - \frac{32}{3} = \frac{2}{3}(9x^2 - 60x + 100) - \frac{32}{3}$

$= 6x^2 - 40x + 56 = 6x^2 - 12x - 28x + 56$

$= 6x(x - 2) - 28(x - 2) = (x - 2)(6x - 28)$

$= 2(x - 2)(3x - 14)$

4 a)

$x^2 - x = 0 \Rightarrow x(x - 1) = 0$ so $x = 0$ or $x = 1$

4 b)

$x^2 - 4 = 0 \Rightarrow (x - 2)(x + 2) = 0$ so $x = 2$ or $x = -2$

4 c)

$x^2 - 8 = 0 \Rightarrow (x - \sqrt{8})(x + \sqrt{8}) = 0$ so $x = \sqrt{8} = 2\sqrt{2}$ or $x = -\sqrt{8} = -2\sqrt{2}$

4 d)

$x^2 + 2x = 0 \Rightarrow x(x + 2) = 0$ so $x = 0$ or $x = -2$

4 e)

$(x-5)^2 = 0 \Rightarrow x - 5 = 0$ so $x = 5$

4 f)

$x^2 - 12x + 36 = 0 \Rightarrow (x-6)^2 = 0$ so $x = 6$

4 g)

$\frac{1}{5}x^2 - \frac{1}{5}x = 4 \Rightarrow x^2 - x = 20$

$\Rightarrow x^2 - x - 20 = 0 \Rightarrow (x-5)(x+4) = 0$

So $x = 5$ or $x = -4$

4 h)

$(x+3)^2 = 6 \Rightarrow x + 3 = \pm\sqrt{6}$

So $x = -3 \pm \sqrt{6}$

4 i)

$(x-1)(x-3) = 8 - 2x$

$\Rightarrow x^2 - 4x + 3 = 8 - 2x \Rightarrow x^2 - 2x - 5 = 0$

$\Rightarrow x = \frac{2}{2} \pm \frac{1}{2}\sqrt{4+20} = 1 \pm \sqrt{6}$

5 a)

$x = \frac{8 \pm \sqrt{64-20}}{2} = 4 \pm \sqrt{11}$

5 b)

$2x^2 = 40 \Rightarrow x^2 = 20$ so $x = \pm\sqrt{20} = \pm 2\sqrt{5}$

5 c)

$x = \frac{-16 \pm \sqrt{256-192}}{4} = -4 \pm 2$

so $x = -6$ or $x = -2$

5 d)

$x = \frac{5 \pm \sqrt{37}}{6} = \frac{5}{6} \pm \frac{\sqrt{37}}{6}$

5 e)

$x = \pm\sqrt{\frac{b}{a}}$

5 f)

$ax^2 - bx = 0 \Rightarrow (ax - b)x = 0 \Rightarrow x = 0$ or $x = \frac{b}{a}$

6)

Student 1 spotted a solution but didn't spot the second one. Student 2 removed the solution $x = 5$ when dividing through by $(x-5)$.

$(x-5)^2 - (x-5) = 0$

$\Rightarrow (x-5)[(x-5)-1]=0$

$\Rightarrow (x-5)(x-6)=0$

$\therefore x=5$ or $x=6$

7)

$x-3=2\sqrt{x} \Rightarrow (x-3)^2=4x$

$\Rightarrow x^2-6x+9-4x=0 \Rightarrow x^2-10x+9=0$

$x=\frac{10\pm\sqrt{100-36}}{2}=\frac{10\pm 8}{2}$ so $x=9$ or $x=1$

But $x\neq 1$ since $-2\neq 2$

So $x=9$

8)

$2x+2=5\sqrt{x} \Rightarrow (2x+2)^2=25x$

$\Rightarrow 4x^2+8x+4-25x=0 \Rightarrow 4x^2-17x+4=0$

$x=\frac{17\pm\sqrt{289-64}}{8}=\frac{17\pm 15}{8}$ so $x=4$ or $x=\frac{1}{4}$

9)

$2(x^2)^2-11x^2+5=0 \Rightarrow 2p^2-11p+5=0$

So $p=\frac{11\pm\sqrt{121-40}}{4}=\frac{11\pm 9}{4} \Rightarrow p=5$ or $p=\frac{1}{2}$

$p=5$ gives $x=\pm\sqrt{5}$ and $p=\frac{1}{2}$ gives $x=\pm\frac{1}{\sqrt{2}}$

10)

$3x^3+19=\frac{40}{x^3} \Rightarrow 3p+19=\frac{40}{p}$

$\Rightarrow 3p^2+19p-40=0$

So $p=\frac{-19\pm\sqrt{361+480}}{6}=\frac{-19\pm 29}{6} \Rightarrow p=\frac{5}{3}$ or $p=-8$

Which gives $x=\sqrt[3]{\frac{5}{3}}$ or $x=\sqrt[3]{-8}=-2$

11)

$12x^4-12x^3+5x^2-x-2=0$

$(2x^2-x)(2x^2-x)=4x^4-4x^3+x^2$

$3(4x^4-4x^3+x^2)+2x^2-x-2=0$

$3p^2+p-2=0$

$\Rightarrow p=\frac{-1\pm\sqrt{25}}{6}$ so $p=-1$ or $p=\frac{2}{3}$

So $-1=2x^2-x$ gives $2x^2-x+1=0$

$x=\frac{1\pm\sqrt{1-8}}{4}$ which gives no real solutions

So $\frac{2}{3}=2x^2-x$ gives $6x^2-3x-2=0$

$x=\frac{3\pm\sqrt{9+48}}{12}$ so $x=\frac{1}{4}\pm\frac{\sqrt{57}}{12}$

12 a)

$x^2 - 8x = (x-4)^2 - 4^2 = (x-4)^2 - 16$

12 b)

$x^2 + 6x = (x+3)^2 - 3^2 = (x+3)^2 - 9$

12 c)

$x^2 - 10x + 1 = (x-5)^2 - 5^2 + 40$

$= (x-5)^2 + 15$

12 d)

$x^2 - 8x - 20 = (x-4)^2 - 4^2 - 20$

$= (x+4)^2 - 36$

12 e)

$x^2 + 5x + 7 = (x + \frac{5}{2})^2 - \frac{5^2}{2} + 7$

$= (x + \frac{5}{2})^2 + \frac{3}{4}$

12 f)

$2x^2 - 16x - 20 = 2(x^2 - 8x) - 20$

$= 2((x-4)^2 - 4^2) - 20$

$= 2(x-4)^2 - 2 \times 16 - 20$

$= 2(x-4)^2 - 52$

12 g)

$4x^2 - 24x + 25 = 4(x^2 - 6x) + 25$

$= 4((x-3)^2 - 3^2) + 25$

$= 4(x-3)^2 - 36 + 25 = 4(x-3)^2 - 11$

12 h)

$3x^2 - 7x + 1 = 3(x^2 - \frac{7}{3}x) + 1$

$= 3((x - \frac{7}{6})^2 - \frac{7^2}{6}) + 1$

$= 3(x - \frac{7}{6})^2 - 3 \times \frac{49}{36} + 1$

$= 3(x - \frac{7}{6})^2 - \frac{37}{12}$

13 a)

$-x^2 + 2x = -(x^2 - 2x)$

$= -((x+1)^2 - 1^2) = 1 - (x+1)^2$

13 b)

$(x+3)(x+2) = x^2 + 5x + 6$

$= (x + \frac{5}{2})^2 - (\frac{5}{2})^2 + 6 = (x + \frac{5}{2})^2 - \frac{1}{4}$

13 c)

$-2x^2 - 4x + 5 = -2(x^2 + 2x) + 5$

$= -2((x+1)^2 - 1^2) + 5$

$= -2(x+1)^2 + 2 \times 1^2 + 5$

$= -2(x+1)^2 + 7$

13 d)

$-2x^2 - 7x + 1 = -2(x^2 + \frac{7}{2}x) + 1$

$= -2((x + \frac{7}{4})^2 - (\frac{7}{4})^2) + 1$

$= -2(x + \frac{7}{4})^2 + 2 \times \frac{49}{16} + 1$

$= -2(x + \frac{7}{4})^2 + \frac{57}{8}$

13 e)

$-3x^2 + 5x + 10 = -3(x^2 - \frac{5}{3}x) + 10$

$= -3((x - \frac{5}{6})^2 - (\frac{5}{6})^2) + 10$

$= -3(x - \frac{5}{6})^2 + 3 \times \frac{25}{36} + 10$

$= -3(x - \frac{5}{6})^2 + \frac{145}{12}$

13 f)

$-4x^2 + 25$ is already in completed square form

13 g)

$x^2 + px + r = (x + \frac{p}{2})^2 - (\frac{p}{2})^2 + r$

$= (x + \frac{p}{2})^2 - \frac{p^2}{4} + r$

14)

(a, b) is the coordinate of the minimum of the curve

15 a)

Minimum at $(2, -12)$

15 b)

Minimum at $(-2, -8)$

15 c)

Minimum at $(1, 12)$

15 d)

Minimum at $(-2, 0)$

15 e)

Minimum at $(-4, 5)$

15 f)

Minimum at $(-\frac{5}{2}, \frac{1}{2})$

15 g)

Minimum at $(4, -100)$

15 h)

$x^2 - 6x - 2$

$= (x-3)^2 - 3^2 - 2$

$= (x-3)^2 - 11$

Minimum at $(3, -11)$

15 i)

$3x^2 + 6x + 5 = 3(x^2 + 2x) + 5$

$= 3((x+1)^2 - 1^2) + 5$

$= 3(x+1)^2 - 3 \times 1^2 + 5$

$= 3(x+1)^2 + 2$

Minimum at $(-1, 2)$

15 j)

$-2x^2 - 4x + 3 = -2(x^2 + 2x) + 3$

$= -2((x+1)^2 - 1^2) + 3$

$= -2(x+1)^2 + 2 + 3$

$= -2(x+1)^2 + 5$

Maximum at $(-1, 5)$

15 k)

Minimum at $(0, -5)$

15 l)

Maximum at $(0, 12)$

15 m)

Minimum at $(b, 0)$

15 n)

$x^2 + px + r = (x + \frac{p}{2})^2 - (\frac{p}{2})^2 + r$

$= (x + \frac{p}{2})^2 - \frac{p^2}{4} + r$

Minimum at $(-\frac{p}{2}, r - \frac{p^2}{4})$

16 a)

$x = 4$

16 b)

$x = -3$

16 c)

$x = 1$

16 d)

Midpoint is $\frac{-4+2}{2}$ so $x = -1$

16 e)

Midpoint is $\frac{1+6}{2}$ so $x = \frac{7}{2}$

16 f)

Midpoint is $\frac{-2+8}{2}$ so $x = 3$

16 g)

Midpoint is $\frac{a+b}{2}$ so $x = \frac{a+b}{2}$

16 h)

$x = p$

17 a)

$k(x-4)(x+1) = k(x^2 - 3x - 4)$ so $k = 1$

$y = x^2 - 3x - 4$

$y = (x - \frac{3}{2})^2 - \frac{3^2}{2} - 4 = (x - \frac{3}{2})^2 - \frac{25}{4}$

Vertex at $(\frac{3}{2}, -\frac{25}{4})$

17 b)

$k(x+2)(x-5) = k(x^2 - 3x - 10)$ so $k = -1$

$y = -(x+2)(x-5)$

Midpoint is at $\frac{-2+5}{2}$ so $x = \frac{3}{2}$ and $y = \frac{49}{4}$

Vertex at $(\frac{3}{2}, \frac{49}{4})$

18 a)

$y = a(x-2)^2 + 2$

At $(0, 6)$; $6 = a(0-2)^2 + 2 \therefore a = 1$

$y = (x-2)^2 + 2$

18 b)

$y = a(x+1)^2 + 8$

At $(0,7)$; $7 = a(0+1)^2 + 8 \therefore a = -1$

$y = -(x+1)^2 + 8$

18 c)

$y = a(x-3)^2 + 5$

At $(0,23)$; $23 = a(0-3)^2 + 5 \therefore a = 2$

$y = 2(x-3)^2 + 5$

18 d)

Midpoint is $\frac{-1+2}{2}$ so $x = \frac{1}{2}$ and $y = -\frac{9}{4}$

Vertex at $(\frac{1}{2}, -\frac{9}{4})$

$y = a(x - \frac{1}{2})^2 - \frac{9}{4}$

At $(0,-2)$: $-2 = a(0 - \frac{1}{2})^2 - \frac{9}{4} \therefore a = 1$

$y = (x - \frac{1}{2})^2 - \frac{9}{4}$

18 e)

Midpoint is $\frac{-1-3}{2}$ so $x = -2$ and $y = 1$

Vertex at $(-2, 1)$

$y = a(x+2)^2 + 1$

At $(0,-3)$: $-3 = a(0+2)^2 + 1 \therefore a = -1$

$y = -(x+2)^2 + 1$

19 a)

$y = a(x-1)^2 + 5$ at $(0,6)$ $6 = a + 5$

So $a = 1 \therefore y = (x-1)^2 + 5$

$y = x^2 - 2x + 6$

19 b)

$y = a(x-3)^2 - 1$ at $(0,17)$ $17 = 9a - 1$

So $a = 2 \therefore y = 2(x-3)^2 - 1$

$y = 2x^2 - 12x + 17$

19 c)

$y = a(x+2)^2 + 4$ at $(0,-76)$ $-76 = 4a + 4$

So $a = -20 \therefore y = -20(x+2)^2 + 4$

$y = -20x^2 - 80x - 76$

20 a)

$2x + 3 = x^2 + x + 1$ gives $x^2 - x - 2 = 0$

So $x = 2$ or $x = -1$

Which gives $(2, 7)$ and $(-1, 1)$

20 b)

$-2x + 3 = x^2 - 2x + 2$ gives $x^2 - 1 = 0$

So $x = 1$ or $x = -1$

Which gives $(1, 1)$ and $(-1, 5)$

20 c)

$2 - x = -x^2 - x + 6$ gives $x^2 - 4 = 0$

So $x = 2$ or $x = -2$

Which gives $(2, 0)$ and $(-2, 4)$

20 d)

$-2x^2 + x + 4 = -x^2 + x + 3$

gives $x^2 - 1 = 0$

So $x = 1$ or $x = -1$

Which gives $(1, 3)$ and $(-1, 1)$

20 e)

$x^2 - 2x = 2x^2 + x - 4$

gives $x^2 + 3x - 4 = 0$

So $x = 1$ or $x = -4$

Which gives $(1, -1)$ and $(-4, 24)$

21 a)

$x^2 + 1 = \frac{1}{2}x + 1$ gives $x^2 - \frac{1}{2}x = 0$

So $x = 0$ or $x = \frac{1}{2}$

Which gives $(0, 1)$ and $(-4, 24)$

21 b)

$-x^2 + 5 = -x + 3$ gives $x^2 - x - 2 = 0$

So $x = 2$ or $x = -1$

Which gives $(2, 1)$ and $(-1, 4)$

21 c)

$x^2 - 4x + 5 = 2x - 4$ gives $x^2 - 6x + 9 = 0$

So $x = 3$

Which gives $(3, 2)$

21 d)

$2x^2 - 4x + 3 = 4x - 5$ gives $2x^2 - 8x + 8 = 0$

So $x = 2$

Which gives $(2, 3)$

22)

Straight line:

$m = \frac{4}{\frac{4}{3}} = 3$ and $c = -4$

$y = 3x - 4$

Quadratic curve:

$y = (x + 4)(x - 3) = x^2 + x - 12$

Solving simultaneously:

$3x - 4 = x^2 + x - 12$ gives $x^2 - 2x - 8 = 0$

$(x - 4)(x + 2) = 0$ so $x = -2$ or $x = 4$

Which gives $(-2, 10)$ and $(4, 8)$

23)

$y = -(x - 2)(3x + 7) = -3x^2 - x + 14$

$y = (x + 1)(x + 6) = x^2 + 7x + 6$

$-3x^2 - x + 14 = x^2 + 7x + 6$

$\Rightarrow 4x^2 + 8x - 8 = 0$

So $x = -1 + \sqrt{3}$ and $x = 1 + \sqrt{3}$

$(-1 + \sqrt{3}, 3 + 5\sqrt{3})$ and $(-1 - \sqrt{3}, 3 - 5\sqrt{3})$